Good dataset and careful processing.

Must fix:

1. Add the modelling workflow. train/test split, CV metric, single test touch and no final refit. Add all five steps from the brief.
2. Encode hr, season, mnth and weathersit as categorical. They are integer codes. A linear or quadratic term in hr cannot capture the two commuter peaks. If you keep them numeric in the baseline on purpose, say so and treat it as the failure to diagnose.
3. Handle time dependence in the split. Adjacent hours are strongly correlated. A random split puts neighbouring hours in train and test and makes the test error optimistic. Use a time-based split for the test set (for example, hold out the final months of 2012). This is a core issue, not an extension.
4. Add VIF. season is a function of mnth, so expect extreme VIF.
5. Plan the influence adjudication. With 17,379 rows, standard cut-oXs will flag hundreds of points. State how you will adjudicate them, for example by grouping flagged hours by date and checking for known events.
6. Partial F-tests are inference. Decisions on which variables or polynomial terms to keep must go through CV on the training set.
7. Added-variable plots are informative for continuous predictors (temp, hum, windspeed). They are not useful for a categorical weathersit.
8. If you model log(cnt), back-transform predictions before computing RMSE on the count scale.

Extension: Pick one. Ridge and LASSO with λ by CV is the natural choice: discuss whether shrinkage addresses the nonlinearity or only the variance. Drop PCA. Every model choice follows the brief's workflow: split once, do all EDA and comparisons with CV inside the training set, state the CV metric before you reveal the winner, touch the test set once, refit on the full data. Add the AI declaration to your report.